

DB9 Rear light permanent condensation fix



Background

It's a common problem to get condensation in the rear light clusters. If left untreated, it leads to permanent damage of the electronic LED driver board inside the light cluster not to mention that the water droplets can damage beyond repair the reflective surface inside the cluster.

I initially noticed and researched the issue at the end of last winter. So far it has only affected the offside cluster but I know its only a matter of time before the other one shows the signs so both needed treating.

After a trip to the local hand car wash, I got multiple light errors on the dash even though the cluster itself was functioning correctly. I guess the loading on the car lighting circuits was incorrect leading to the errors being detected.

I was never convinced that the AM TSB appeared to be a satisfactory and permanent fix so I devised my own plan involving providing permanent access to the light interior and then to use silica gel to ensure that the cluster remain dry on the inside. I wanted the ability to remove and dry or replace the gel if needed later but obviously I wanted the light to remain sealed in the meantime.

I decided it was time to put my plan into action 😊

The Plan

1. Remove the light
2. Disassemble the light by removing the lens
3. Clean and dry the light
4. Cut a 50mm hole in the bottom of the rear case
5. Test fit a rubber grommet to the 50mm hole
6. Replace the lens seal with non-permanent butyl rubber sealant
7. Reassemble the light cluster
8. Place 3 x 20g silica gel sachets inside the light via the 50mm hole and seal with the grommet
9. Refit the light

Simple!!!

At this point I will point out that I did one side in full at a time. This allowed me to check the refit alignment of the repaired cluster against the other light that had not yet been removed.

Step 1-Removing the light

In order to access the light fixings, the carpet must be removed from inside the boot behind the lights. Initially the boot seal needs to be removed to give access to the carpet section behind the lights. The seal just pulls off. The carpet is held in place using double sided tape so the carpet can be peeled away. It does not need to be fully removed.

Once removed, it gives you access to the 3 17mm nuts that hold the light in place



With the nuts, shake proof washers and penny washers removed, the light cluster can be removed away from the car. The light is still plugged into the car at this point.

With the light moved away from the body, you can access the plug. You need to be careful not to let the mounting bolts scratch the bodywork.

When you can see the plug, the red locking mechanism can be undone. If its too tight to move by hand, you can insert a small thin flat plastic trim tool into the slot and lever the locking mechanism away from the plug. As you do this, the plug will release and move away from the cluster.



With the light removed you can proceed to step 2

Step 2 – Light Disassembly

In order to remove the lens, the light must be heated to soften the sealing strip and also the tabs must be warm so they do not break as you bend them away from the body.

I found that the red plastic on the lens goes soft at a much lower temperature than I expected and that the plastic became malleable. I decided to heat and bend the tabs away while the plastic was soft. This way they are not in play and holding the lens in place as you prise the lens away from the body. I can heat and bend the tabs flat again before reassembly.

Once the tabs were safely bent away from their landing on the body, I carefully and gently heated around the seal area. I directed the heat more towards the body than the lens as I learned heating the tabs how easily this can deform.

On previous headlight projects, I've heated the entire light in the oven before I started to pry the lens away. I had learned that repeated heating was required to remove the lens so this time I used a heat gun on a low speed and medium heat.

The trick here is to start to get the cluster apart from one of the corners. The reason is that you don't want to deform the channel in the lens as you pry apart. The corner has the most strength and so you are less likely to damage the lens as you apply the force required to start splitting the cluster.

You need to feel for movement and not be too heavy handed as you pre the lens away from the body. Work along the seal applying more heat as required using pry tools and leave a pry tool in the gap as it opens. I took me about 20 or so cycles with the heat to work my way around the whole cluster.

***Ignore the broken clips in these pics. The pics are staged on an old spare light that was already broken when I got it. I staged the pics as I didn't take any while I was prying my light cluster apart!!!



Once the lens is free, you can remove the light internals.

You will need to be careful not to touch or scratch the reflective surface of the light so I worked with the light face down on a soft surface while I removed internals

The plug on the back is located using 2 tabs either side to hold it in place. These tabs need to be pressed towards the plug itself in order to push the plug into the light through its hole in the body.



There are 3 silver torx bolts that need to be removed to free the light internals



With the bolts removed, you can extract the entire light internals. I pulled the internals out using the aluminium blade that is used to mount the LEDs as not to touch the reflective coating.

Step 3 -Clean and Dry the light

With light disassembled, you can move on the cleaning the light.

I dried the internals using the heat gun and set it aside safely.

There was far more water inside the light than I was expecting so I carefully removed excess using a paper towel. You need to be careful not to apply too much heat to the electronics. You do not want to over heat the capacitors or melt any solder on the PCBs.

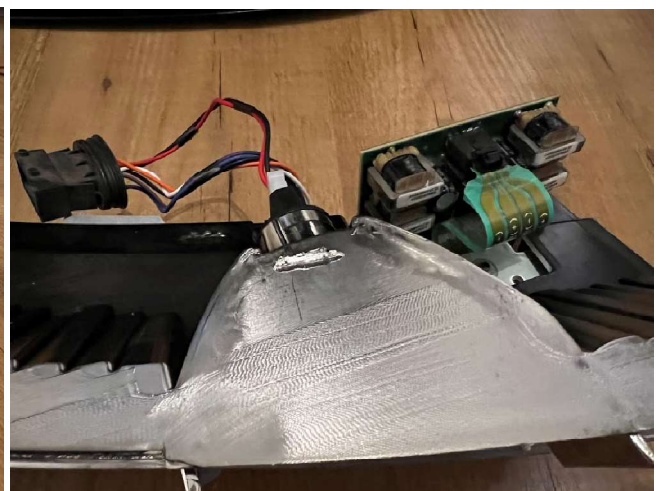
Next you need to clean and dry the inside of the lens. Again I used paper towels initially and then heat to fully dry the lens.

Next you need to remove all of the old sealer strip.

The heat gun was again used to soften the strip and I used a pry tool to get under the sealer and remove it from the lens. You need to be careful not to apply too much heat as you do not want to deform the channel. At this time I also heated the tabs and returned them to their flat position so that they would locate on the rear housing correctly during reassembly.



Finally, I cleaned and dried the rear housing, including the landing for the plug as well as the rubber seal on the plug itself.



With the light fully disassembled and cleaned you can move onto the step 4

Step 4 -Cut a 50mm hole in the bottom of the rear case

Having noted the position that the internal occupy inside the rear housing, I worked out where to drill the 50mm hole. I wanted the hole as central as possible and on a flat part of the housing so that the grommet would seal correctly. I also wanted to ensure that I could relatively easily fit and remove the silica gel sachets.

With an appropriate position identified, I drilled the hole and deburred the edge.



Step 5 – Test fit the grommet

I test fitted the grommet which I already knew was a good fit as I previously tested the fit on some plastic sheet 😊



I also made sure that the grommet on the inside did not interfere with the reassembly of the light internals!

Step 6 – replacement butyl rubber seal

The seal strip that I have is very soft not dissimilar to bluTak in its softness.

I stretched the sealing to approximately twice its original length to get the thickness to something that was appropriate for the size of the channel and laid it into the channel.



Step 7 -Reassembly

Refit the internals including locating the plug back into the housing and replacing the 3 torx bolts to hold it together.

When I refitted the plug into the housing, I used a pick to pull the tabs in the plug out a seat the connector correctly back into the housing.

At this point, you need to ensure that the light is fully dry. You don't want any dampness inside the cluster as you reassemble. Again, I used the heat gun to raise the temperature of all the cluster components to ensure that it was dry and warm for final fitting of the lens. I also made sure that there were no marks on the reflector at this point.

Now you can refit the lens. It was warm so the butyl seal is softer and this helps it mould to the correct shape as you refit the lens.

With the light cluster on a hard surface facing up, I alight and pressed home the lens working around the cluster until all 5 tabs had located correctly.

You are now left with a freshly sealed cluster with a 50mm hole in the bottom!

Step 8 -Place 3 x 20g silica gel sachets inside the light via the 50mm hole and seal with the grommet

You can now install the silica gel through the hole 😊

I used 3 rechargeable 20g sachets pushing them to the left, right and just above the hole.

The grommet was then fitted ensuring that it seats correctly which meant working the edge of the grommet until it was perfectly flat.



Step 9 -refit the cluster to the car

Refit is the reverse of removal.

If the adjusting collars on the stud of the cluster have moved during disassembly, you will need to re align them by checking the shut lines around the light against the body work of the car. Doing one side at a time allowed me to compare the refitted light against the factory fit.

Once aligned, the 17mm nuts and washers can be fitted to hold the light in place and carpet refitted.

Once both sides are completed, the boot seal can be refitted.



Parts List

[Butyl Rubber Seal – eBay](#)

[Rubber Grommet 50mm – Amazon](#)

[Silica Gel 20g – Amazon](#)

Conclusion

In theory, this is better solution than Aston came up with but of course it's a fair bit more work.

The lights have been back on the car for a week or so in some pretty wet weather here in the UK. So far, no condensation has returned. I'm happy that the cluster resealed and I now know it's a far more straightforward job if I need to dry and recharge the silica gel sachets in future.

I now also plan to drop some similar sachets of silica gel into the headlights to resolve the same condensation issue with those. I think I should be able to drop them in through the bulb access covers making the job far much easier!

Hope you find this useful! Ian B 2022